

Wireless Security Lab: Smartphones Spoofing

Colombino, F. (s349461) Papantuono, F. (s360951) Pedron, M. (s285048) Stella, P. (s364246)
Politecnico di Torino

ABSTRACT

This laboratory evaluates GNSS spoofing effects on Android smartphones across open-space, RF-interference, and urban multipath environments using a software-injected attacks simulating coordinate displacement and time delays.

1 METHODOLOGY AND EXPERIMENTAL DESIGN

1.1 Data Collection Infrastructure

1.1.1 Hardware Specifications. The experimental campaign utilized three distinct Android smartphones to ensure diversity in chipset architecture and GNSS firmware behavior.

- **Smartphone A:** Asus ZenFone 4 Max; OS: Android 8.1.0; Chipset: Qualcomm Snapdragon 430.
- **Smartphone B:** Xiaomi Redmi Note 10; OS: Android 12; Chipset: MediaTek Helio G95.
- **Smartphone C:** OnePlus Nord 2T; OS: Android 14; Chipset: MediaTek Dimensity 1300.

1.1.2 Software Compatibility and Versioning. All datasets were logged using GNSSLogger v3.1.1.2, available from the official Google repository.

1.2 Experimental Environments and Procedures

1.2.1 Environmental Baseline: OpenSpace Dataset. A reference baseline under nominal GNSS conditions established using Smartphone A in the area above Politecnico Rooms I.¹ This location provides an unobstructed hemispherical view of the sky with optimal satellite geometry and minimal signal attenuation. The resulting data serve as a control group for evaluating the degradation effects observed in more complex urban or high-interference scenarios.

1.2.2 Radio-Frequency Interference: Maddalena Dataset. Smartphone B was deployed at Colle della Maddalena², where dense broadcast infrastructure served to evaluate if adjacent-band emissions desensitize the GNSS receiver's low-noise amplifier (LNA) or elevate the noise floor, as reflected in C/N_0 distributions.

1.2.3 Multipath and Obstruction: Dumarey Building Dataset. To analyze near-field obstruction and Non-Line-of-Sight (NLoS) impacts, Smartphone C logged static, ground-level GNSS data at the Dumarey facility³ (Politecnico di Torino). Positioned near the facade, this proximity physically obstructed a vast sector of the sky dome, causing severe azimuthal masking and inflating GDOP. This structure forced NLoS signals to bounce, strongly emphasizing

¹<https://maps.app.goo.gl/3r5E822u71dQwHqw7>

²<https://maps.app.goo.gl/6ASxJNceVYFjbe8x6>

³<https://maps.app.goo.gl/si8SUPk2TyUSSXdZA>

multipath phenomena and inducing significant geometric delays that likely drove the inflated pseudorange variance and systematic biases subsequently observed in the GpsWlsPvt estimation.

1.2.4 Power Management Constraints. A fourth measurement was deployed using Smartphone B with power saving enabled to investigate on modifications in HardwareClockDiscontinuityCount. There were no immediate discontinuities, suggesting that Smartphone B's chipset may resist clock duty-cycling during these specific logging durations, though further investigation is required for generalization.

1.3 Analytical Processing Pipeline

The processing pipeline employs a modified version of van Diggelen's open-source *GPS Measurement Tools* [4], extended by the NavSAS Research Group with spoofing simulation capabilities. The workflow proceeds as follows: ReadGnssLogger parses raw logs applying filters; GetNasaHourlyEphemeris retrieves precise ephemerides from NASA CDDIS; ProcessGnssMeas converts raw counts to pseudoranges, integrating synthetic ranges via compute_spoofSatRanges when spoof.active=1; and GpsWlsPvt computes the Weighted Least Squares position-velocity-time solution. The pipeline generates six diagnostic outputs (pseudoranges, C/N_0 , PVT trajectories, residuals).

1.4 Spoofing Simulation Parameters

Cyberspoofing was emulated via MATLAB manipulation of spoof.position and spoof.delay. Authentic pseudoranges were replaced with geographically plausible yet statistically significant counterfeit values. A 15 s pre-attack window facilitated comparative analysis between the stable baseline and induced instability. The applied spoof.delay (1×10^{-3} s) emulated meaconing-induced propagation latency, evaluating WLS clock bias resilience against uniform pseudorange inflation.

2 RESULTS AND DISCUSSION

2.1 MATLAB Plot Outputs

Pseudorange Change from Initial Value shows range variations normalized to zero, producing a "scissor" pattern as satellites approach or recede. Hardware Clock Discontinuity detects clock resets from power management that introduce pseudorange step-functions. Pseudorange Rate ($\text{diff}(\text{PR})/\text{diff}(\text{time})$) shows the discrete-time derivative of pseudoranges; abrupt variations indicate spoofing or tracking anomalies. C/N_0 displays signal power for each satellite; concurrent attenuation across channels indicates RF interference. Raw Pseudoranges plots horizontal position with a circle centered at the median containing 50% of positions (CEP radius). Horizontal Speed estimates velocity magnitude; large deviations from zero (for static receivers) indicate outliers. HDOP quantifies geometric dilution; lower values indicate better accuracy. The orange trace shows satellite count. Common Bias Clock

Offset shows receiver clock bias; linear drift indicates oscillator instability.

2.2 Dataset 1 – Open Space

2.2.1 Data Presentation. Initial baseline analysis under open-sky conditions reveals continuous tracking of all visible satellites throughout the 180-second acquisition period (Figure 1). The C/N_0 analysis (Figure 2) exhibits stable signal quality with an overall mean of 29.76 dB-Hz. Specifically, PRN 16 maintains the highest strength at approximately 38 dB-Hz, while lower-elevation satellites demonstrate reduced but stable values ranging between 22–33 dB-Hz, with no observable signal attenuation or tracking instabilities. Furthermore, the pseudorange rate observables (Figure 3) reveal consistently negative derivatives, bounded between -1200 and -2000 m/s for all satellites. The absence of hardware clock discontinuities confirms that power-saving modes were successfully disabled during the logging phase.

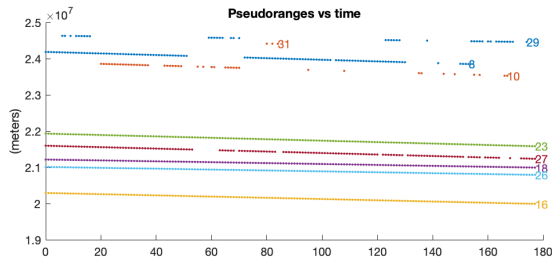


Figure 1: Pseudoranges vs. Time showing continuous tracking of all visible satellites under open-sky conditions.

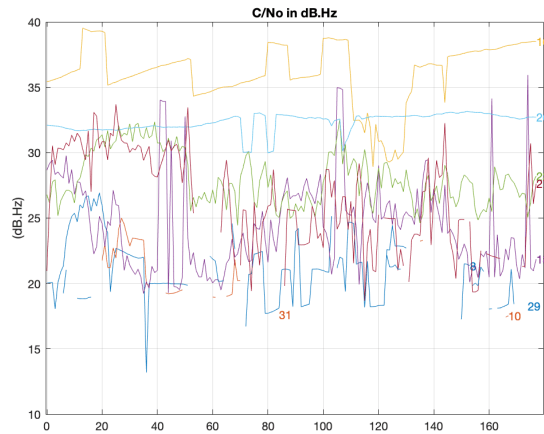


Figure 2: Carrier-to-Noise density (C/N_0) profiles indicating stable signal quality and optimal line-of-sight.

The WLS position solution (Figure 4) yields an estimated median at [45.0651389, 7.6584375, 313.79]. As a result, 50% of computed positions fall within a tightly constrained 4.25 m CEP radius, with a maximum horizontal error of 16.01 m. The altitude varies from 262.26 m to 366.19 m, producing a 103.93 m range and a 20.10 m standard deviation. The spatial distribution shows a distinct East-West anisotropy: the East Variance (27.20 m²) exceeds the North Variance (4.50 m²) by a factor of six. Satellite geometry remains stable, maintaining 4 to 6 tracked satellites continuously. Horizontal

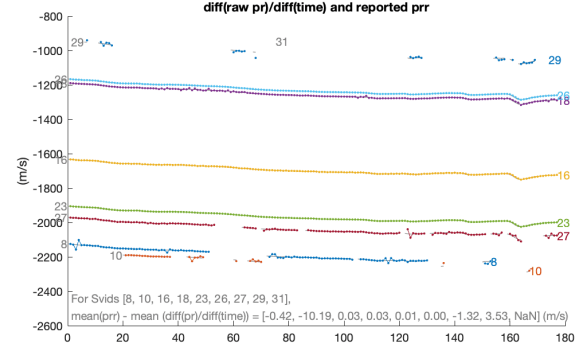


Figure 3: Pseudorange rate observables demonstrating systemic negative derivatives driven by hardware clock drift.

velocity estimates oscillate near zero, punctuated by occasional ± 10 m/s spikes. Finally, the common bias clock offset exhibits a cumulative linear drift of -3×10^5 m (equivalent to a -300 km range) over 180 seconds, which corresponds to a drift rate of -1593.92 m/s.

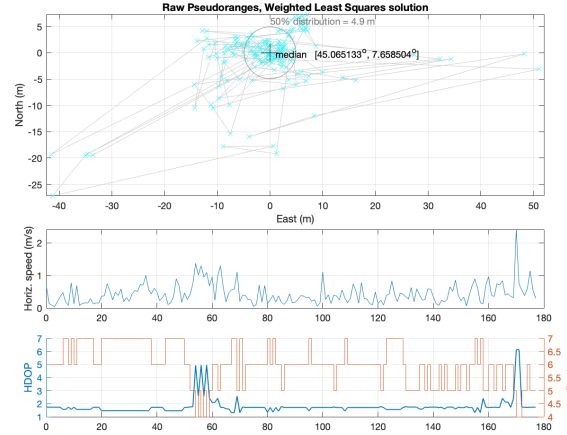


Figure 4: WLS Position Solution revealing high precision (4.25 m CEP) accompanied by distinct East-West spatial anisotropy.

2.2.2 Interpretation. The stable C/N_0 profiles and uninterrupted tracking corroborate that this unobstructed environment provides a highly effective reference baseline, validating nominal receiver performance under optimal conditions. The systematic negative pseudorange rates, however, indicate an uncorrected hardware clock drift rather than genuine satellite convergence. Because orbital mechanics preclude the simultaneous approach of all satellites to a static receiver, this uniform shift reflects inherent mass-market oscillator instability. The WLS filter efficiently absorbs this common-mode timing error into the clock bias state, as mathematically confirmed by the linear -1593.92 m/s drift rate.

Moreover, while the 4.25 m CEP demonstrates achievable precision under clear-sky conditions, the marked East-West anisotropy highlights residual geometric dilution along the longitudinal axis. Similarly, the 103.93 m altitude range reflects typical vertical dilution inherent in consumer-grade GNSS receivers, where geometric constraints naturally degrade altitude accuracy relative to horizontal positioning. The observed velocity spikes represent static receiver noise amplified by discrete-time differentiation rather than actual physical motion.

2.2.3 Limitations. A primary limitation of this dataset is the systematic hardware clock drift affecting all pseudorange rates. While the WLS algorithm successfully absorbs this common-mode error into the user clock bias state, the artifact completely obscures the true radial velocity signatures of individual satellites, effectively preventing the independent validation of Doppler-based velocity estimation. Furthermore, the East-West positional anisotropy, though moderate when compared to constrained environments, indicates that satellite constellation geometry introduces directional bias even under optimal open-sky conditions. Consequently, these specific geometric characteristics may not reliably generalize to other temporal windows or geographic locations.

2.3 Dataset 2 – Maddalena

2.3.1 Data Presentation. The second experimental campaign was conducted near Parco della Maddalena (Turin), an environment characterized by the presence of a high-power radio broadcasting antenna. This site was specifically selected to evaluate the resilience of raw GNSS observables against electromagnetic interference (EMI). Visual inspection of the "Pseudoranges vs. time" plot (Figure 5) confirms significant tracking instability; the receiver frequently fails to maintain a continuous phase lock, resulting in fragmented signal trajectories. Despite these anomalies, the "Pseudoranges change from initial value" analysis preserves a clear "scissor" signature, accurately reflecting the relative orbital kinematics as satellites converge and recede relative to the static receiver.

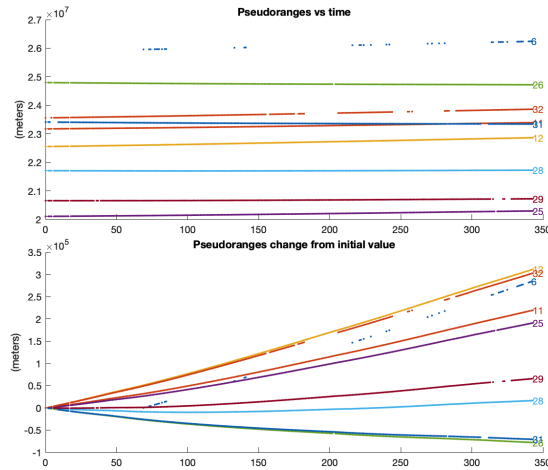


Figure 5: Pseudoranges vs. Time for the Maddalena dataset, showing fragmented trajectories due to local RF interference while maintaining the characteristic orbital kinematics signature.

The impact of the interference is further quantified by the C/N_0 analysis (Figure 6), which reveals five distinct temporal epochs of concurrent signal power attenuation across multiple channels. During these severe degradation events, the receiver struggles to demodulate incident signals, leading to the loss of 2 to 3 satellites from the position fix. Consequently, the Weighted Least Squares (WLS) solution metrics (Figure 7) exhibit notable fluctuations in satellite availability, with the count periodically dropping from a peak of 9 to a minimum of 6. Despite these hostile RF conditions, the system maintains an average tracking of 8 satellites, which

preserves favorable Horizontal Dilution of Precision (HDOP) values, predominantly bounded between 1.5 and 2.5.



Figure 6: Carrier-to-Noise density (C/N_0) profiles highlighting five concurrent attenuation events caused by high-power radio broadcasting emissions.

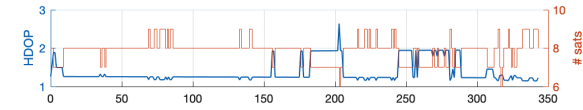


Figure 7: WLS Position metrics indicating fluctuations in satellite count and the corresponding stability of HDOP values despite interference.

Subsequently, a simulated location spoofing attack was injected starting at $t = 15$ s, targeting a spatial offset of approximately 100 m from the true coordinates at [45.0313399, 7.7213363, 776.91]. The transition is immediately evident in the positioning domain (Figure 8), where the spatial trajectory is abruptly translated to the counterfeit coordinates. While the initial samples accurately depict the true location, all epochs following the 15-second mark conform strictly to the spoofed path. In the pseudorange rate (PRR) domain, the attack induces severe noise and erratic patterns, transitioning from nominal linear behavior to extreme numerical discontinuities.

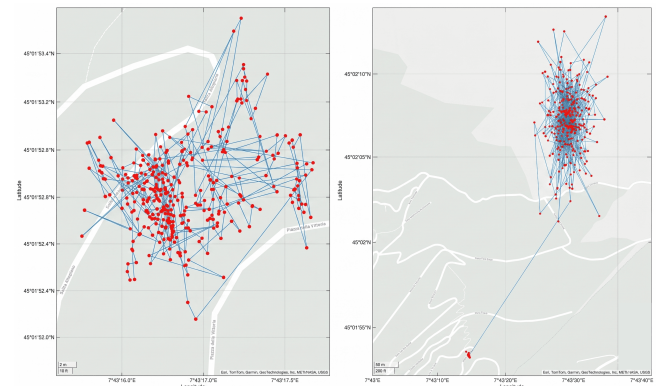


Figure 8: Spoofed spatial trajectory showing the instantaneous coordinate translation at $t = 15$ s towards the target counterfeit position.

2.3.2 Interpretation. The observed tracking instability and signal dropouts are directly attributable to the high-power electromagnetic emissions from the adjacent antenna. The precise correlation between C/N_0 attenuation epochs and satellite dropouts confirms that RF interference desensitizes the receiver's front-end, elevating the noise floor and disrupting tracking loops. However, the preservation of the "scissor" pattern and the maintained HDOP range (1.5–2.5) demonstrate a degree of geometric resilience; the receiver sustains basic positioning capability even when the signal-to-noise ratio is severely compromised.

The catastrophic perturbations in the PRR observables following the spoofing injection are a direct mathematical consequence of the discontinuous spatial translation. Because the pseudorange rates are computed as the discrete time derivative $\Delta\rho/\Delta t$, the instantaneous coordinate jump commanded by the spoofing parameters generates massive numerical spikes. This behavior provides an unambiguous signature for attack detection, clearly distinguishing algorithmic manipulation from nominal orbital motion.

2.3.3 Limitations. A primary limitation of this analysis is the reliance on post-processing software injection to emulate the spoofing attack. While this method successfully validates the WLS estimator's response to counterfeit observables, it does not replicate hardware-level dynamics such as Automatic Gain Control (AGC) variations or authentic correlation peak distortions that would occur during an over-the-air RF event. Additionally, the lack of quantitative measurements regarding the interfering antenna's exact transmission power and spectral characteristics prevents the establishment of a rigorous mathematical correlation between the interference power and the observed C/N_0 degradation.

2.4 Dataset 3 – Dumarey

2.4.1 Data Presentation. Initial baseline signal analysis reveals a significant degradation in overall quality. Specifically, the C/N_0 of PRN 15 exhibits a highly periodic fluctuation (17–27 dB-Hz) before repeatedly losing lock. While the exact physical mechanism remains undetermined, this sawtooth pattern strongly suggests severe multipath fading, where constructive and destructive reflections cyclically disrupt the receiver's tracking loops. Regardless of the exact origin, this signal instability directly correlates with cyclic variations in the estimated altitude (Figure 9). Consequently, the altitude produced a massive 321.88 m range (Table 1, Baseline), shifting the median to 476.71 m—a severe vertical bias when compared to the 249 m true topographical altitude.

Furthermore, HDOP metrics highlight poor satellite geometry (fluctuating between 1.5 and 3.5), exhibiting perfect inverse proportionality to the number of tracked satellites, which averaged only 5.5. As a result, 50% of computed positions fall within a broad 44.6 m CEP radius (Table 1, Baseline), centered at [45.063668, 7.658522] (Figure 10). This dispersion is heavily driven by spatial outliers reaching a maximum distance from the mean of 220.79 m. Visually, the spatial distribution forms a straight line strictly perpendicular to the facade. This geometric distortion is mathematically confirmed by the directional variances: the East Variance is an order of magnitude larger than the North Variance, as shown in Table 1. The underlying cause of this distribution is discussed below.

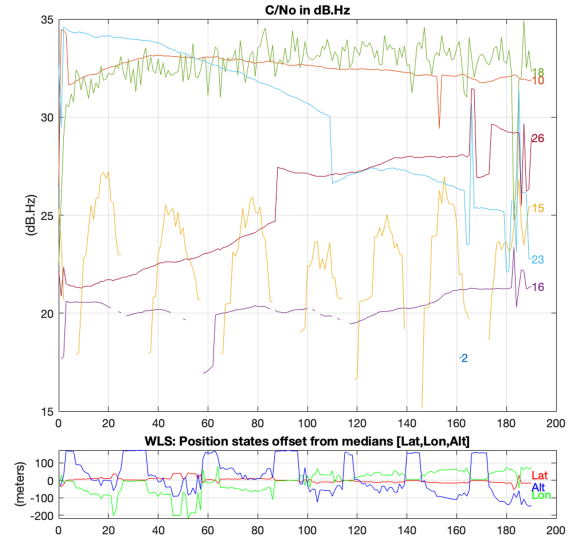


Figure 9: Estimated altitude variations for the Dumarey baseline, highlighting the severe vertical bias and cyclic anomalies caused by multipath fading.

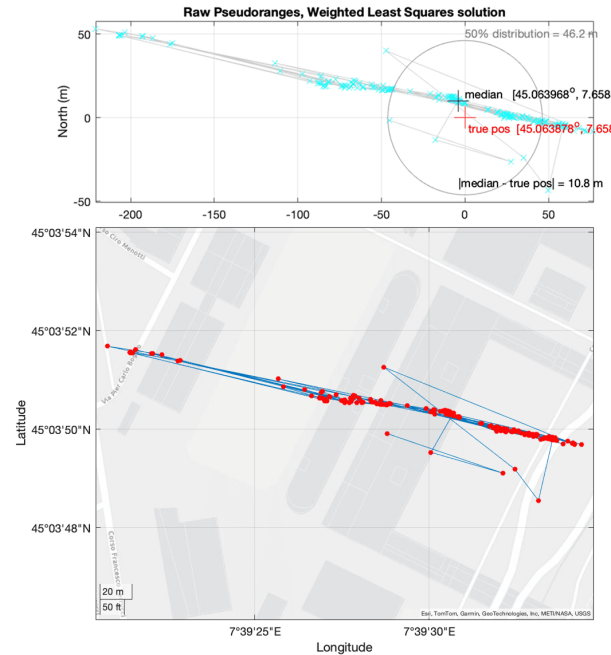


Figure 10: Baseline WLS position solution demonstrating orthogonal, straight-line spatial dispersion perpendicular to the building facade.

A programmatic data filter was implemented to exclude observations below 20 dB-Hz and remove erratic satellites (namely PRNs 2, 15, and 16). Eliminating these noise-dominated signals successfully stabilized the WLS altitude, reducing its error range to 42.69 m and eradicating the periodic artifacts. In this filtered solution, the 50% CEP collapsed to a tightly constrained 4.48 m radius (Table 1, Filtered), representing a full order of magnitude reduction compared to the baseline (Figure 11).

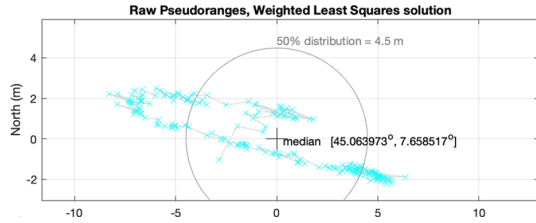


Figure 11: Filtered WLS position solution showing a tightly constrained 4.48 m CEP radius after excluding noise-dominated and erratic signals.

Subsequently, a delay spoofing attack was emulated on the filtered dataset. The attack shifted the estimated median to [45.0639782, 7.6586818, 301.951] (Figure 12) and drastically expanded the 50% CEP to 122.22 m (Table 1, Spoofed). Although the linear distribution persisted, outliers reached a 554.40 m maximum horizontal error. Moreover, latitude and longitude deviations spiked to roughly ± 500 m, while altitude variations exploded to a 3200.36 m range (Table 1, Spoofed). Despite this severe spatial disruption, the attack's onset remains trivially detectable via the PRR derivatives, which mathematically diverge precisely at the 15-second mark.

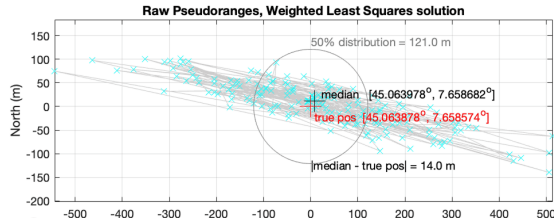


Figure 12: Spoofed spatial trajectory illustrating the catastrophic PVT degradation and massive horizontal error expansion following the 1 ms delay attack.

Table 1: Comparison of WLS Position Metrics across Baseline, Filtered, and Spoofed conditions at the Dumarey site.

Parameter	Baseline	Filtered	Spoofed
Alt. Std. Dev. (m)	98.02	4.64	467.48
Alt. Maximum (m)	625.25	341.87	2141.39
Alt. Minimum (m)	303.37	299.18	-1058.98
Alt. Range (m)	321.88	42.69	3200.36
50% CEP (m)	44.60	4.48	122.22
Horiz. Std. Dev. (m)	66.23	4.64	191.21
East Variance (m ²)	4177.05	19.18	34309.82
North Variance (m ²)	209.25	2.39	2252.29
Max. Horiz. Error (m)	220.79	8.59	554.40

2.4.2 Interpretation. The data robustly illustrates GNSS vulnerability in constrained environments. The orthogonal, straight-line dispersion of the baseline positions is the direct result of poor cross-track geometry combined with severe multipath. With half the sky obscured, the receiver completely lacks transversal spatial diversity. Therefore, it is highly probable that the Dumarey facade reflected NLoS signals, artificially lengthening their pseudoranges and forcing the WLS estimator to stretch the solution along the single, geometrically constrained East-West axis.

Under the spoofing attack, the catastrophic PVT degradation highlights the system's critical sensitivity to timing manipulation.

A 1 ms artificial delay injected into the tracking loops translates instantaneously into a 300 km synthetic range extension. Without RAIM to isolate the outlier, this discontinuous pseudorange jump forces the navigation filter to diverge, completely hijacking the PVT solution.

Additionally, a minor, global C/N_0 attenuation observed at the log's end likely resulted from human-body path loss. This perturbation occurred inadvertently when the operator physically leaned over the smartphone's internal antenna.

2.4.3 Limitations. A significant anomaly in the PRR observables showed only satellites presenting a negative slope. While the varying negative gradients correctly imply different radial velocities, it is highly improbable from an orbital mechanics perspective that all tracked satellites were physically converging on the static receiver simultaneously (Figure 13). Indeed, verification of the satellite trajectories over the test area using a tracking satellites tool confirmed this thesis. A more rigorous hypothesis points to an uncorrected hardware clock drift within the mass-market oscillator. Because the WLS navigation filter natively absorbs common-mode timing errors into the user clock bias state, this hardware frequency drift maps directly into the observables, uniformly shifting all PRR derivatives downward and generating the illusion of a systemic convergence.

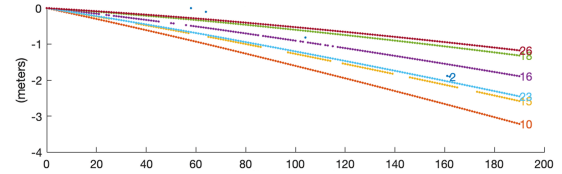


Figure 13: Pseudorange rate observables exhibiting a uniform negative slope across all tracked satellites, indicative of uncorrected hardware clock drift.

3 CONCLUSION

This study evaluated mass-market Android GNSS receiver resilience against environmental and adversarial interference. Severe multipath and NLoS reflections compromised accuracy by introducing substantial spatial variance. While C/N_0 -based filtering successfully stabilized the PVT solution against these artifacts, receivers remained highly vulnerable to intentional attacks. High-power RF interference desensitized the front-end, causing widespread tracking instabilities. Furthermore, algorithmic delay injections bypassed the WLS filter, instantaneously hijacking the navigation solution and exposing inherent susceptibilities to cyberspoofing.

A primary limitation remains the reliance on software-injected spoofing rather than over-the-air RF transmissions. Although this approach validates estimator failure and PRR divergences, it inherently bypasses hardware-level AGC variations essential for genuine interference detection. Future research should prioritize physical RF spoofing simulations to rigorously evaluate integrated AGC and C/N_0 cross-checking countermeasures.

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